Mahalakshmi S

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Professional Summary

Electronics and Communication Engineering student at VIT Vellore with hands-on experience in embedded systems, sensor interfacing. Completed an embedded system design internship at Maven Silicon and built three hardware projects: an offline CNN-based sign language recognizer on Raspberry Pi 5 and ESP32-CAM (94-95% test accuracy), an SPI-based RFID motor controller, and a 4-lane smart traffic signal controller documented in an IEEE-format paper. Seeking an Embedded Systems / IoT role to apply microcontroller programming and hardware-software integration skills to real-world products .

Education

Vellore Institute of Technology , Vellore, India — **Aug 2023 – May 2027(Expected)**
B. Tech Electronics and Communication Engineering — CGPA : **8.96/10**

Government Girls Higher Secondary School, Sholavandan, Tamil Nadu, India — **Jun 2022- May 2023**
Percentage : **94.5 %**

Projects

Offline Sign Language Recognition Using CNN Raspberry Pi 5 | ESP32-CAM | Python | TensorFlow Lite | OpenCV

- Built an edge-AI gesture recognition system in which an ESP32-CAM (OV2640) captures 240×240 hand images and sends them over Wi-Fi (HTTP) to a Raspberry Pi 5, which runs the CNN locally with no cloud or internet dependency for inference.
- Trained a CNN in TensorFlow/Keras on 128×128 RGB gesture images, achieving 94-95% test accuracy (95-96% training, 97-98% validation).
- Quantized the model to INT8 and deployed it with TensorFlow Lite (LiteRT) on the Raspberry Pi 5 (~6.3 MB model), displaying the predicted gesture and confidence score in about 12 s end-to-end from capture to output.

RFID-Based Motor Control System Using Arduino UNO (ATmega328P, 16 MHz) | MFRC522 RFID (13.56 MHz) | SPI | Relay Module | Embedded C

- Built an RFID motor controller on an Arduino UNO (16 MHz) with a 13.56 MHz MFRC522 reader over SPI (5 lines), authenticating tag UIDs against 2 registered UIDs.
- Implemented 2-tag logic (1 tag = motor ON, 1 tag = motor OFF, all other UIDs rejected) with Serial Monitor status (9600 baud, 500 ms scan cycle).
- Switched the motor via a 5V relay on 1 GPIO (D8) with a separate 9V supply from the Arduino's 9V, and tested the 3.3V RFID, 5V relay, 9V motor circuit with authorized and unauthorized cards

Smart Traffic Signal Controller with Density & Emergency Detection Arduino UNO | IR/Ultrasonic Sensors | RF Module | Embedded C | Sensor Interfacing

- Built a 4-lane junction prototype on 1 Arduino UNO, using 2 sensor types (IR + ultrasonic) to estimate lane density and adjust green-signal time.
- Implemented RF-based emergency detection (1 transmitter, 1 receiver) that overrides the cycle, gives the emergency lane an immediate green with a buzzer alert, then restores normal operation.
- Tested the 5V breadboard prototype with red/yellow/green LEDs and documented it in an IEEE-format paper with a 6-member team and 6 references.

Skills

- **Programming Languages:** C, Embedded C, C++(Basics).
- **Microcontrollers & Platforms:** Arduino UNO (ATmega328P), ESP32-CAM, Raspberry Pi 5
- **Embedded & Hardware Interfacing**: GPIO control, relay-based motor switching, RFID (MFRC522, 13.56 MHz), IR and ultrasonic sensor interfacing, RF transmitter/receiver modules, LED and buzzer output control
- **Communication Protocols**: SPI, UART, I2C, CAN
- **Tools & Simulation**: Arduino IDE, Proteus, Tinkercad, Raspberry Pi OS (Linux terminal)
- **Core Concepts**: Embedded systems, Digital and Analog Electronics, Microcontroller Architecture and Programming.

Achievements & Certifications

- 100% Tuition Fee Scholarship under the VIT STARS Scheme (2023–2027) for academic excellence.
- Secured the highest marks among government school students in the district in Higher Secondary Education.
- Embedded System Design Internship: Developed an embedded obstacle detection system for the visually impaired(Link).
- Implementation of ML Algorithms on FPGA – VIT: FPGA-based ML implementation using Verilog HDL(Link).